



# Muhammad Konain Tahir

Computer Science Undergraduate | COMSATS University Islamabad

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## PROFESSIONAL SUMMARY

Computer Science undergraduate (4th semester) at COMSATS University Islamabad, Sahiwal Campus, maintaining a 3.62 CGPA. Experienced in full software development lifecycle through academic projects spanning OOP, database systems, and data structures & algorithms. Proficient in C++, Java, and Python with a strong foundation in problem-solving, software design principles, and algorithmic thinking.

## EDUCATION

### COMSATS University Islamabad, Sahiwal Campus

Bachelor of Science in Computer Science (In Progress)

Sahiwal, Pakistan

Expected 2027

- CGPA: 3.62 / 4.00
- Relevant Coursework: Object-Oriented Programming, Database Systems, Data Structures & Algorithms

## ACADEMIC QUALIFICATIONS

### Higher Secondary Certificate (FSC)

Pre-Engineering / Intermediate

Marks: 765 / 1200 | Grade: B

### Secondary School Certificate (Matric)

Science Group

Marks: 823 / 1100 | Grade: A

## PROJECTS

### Hospital Management System

OOP (Java)

- Developed a comprehensive hospital management application applying core OOP principles including inheritance, polymorphism, and encapsulation.
- Implemented modules for patient registration, doctor scheduling, appointment booking, and billing workflows.

### Airline Management System

Database Systems

- Designed and built a relational database-driven system handling flight scheduling, seat reservations, and passenger records.
- Wrote complex SQL queries for booking management, availability checks, and reporting dashboards.

### Traffic Flow Simulator

Data Structures & Algorithms (C++)

- Built a traffic flow simulation engine leveraging graph-based data structures to model road networks and intersections.
- Implemented pathfinding and scheduling algorithms to simulate real-time vehicle routing and congestion analysis.

## TECHNICAL SKILL

| Languages         | Concepts                 | Tools              |
|-------------------|--------------------------|--------------------|
| C++, Java, Python | OOP, DSA, DB Design, SQL | Git, GitHub, MySQL |